

Asymmetric Cryptographic Techniques - Principles

Module III

Public Key Cryptography

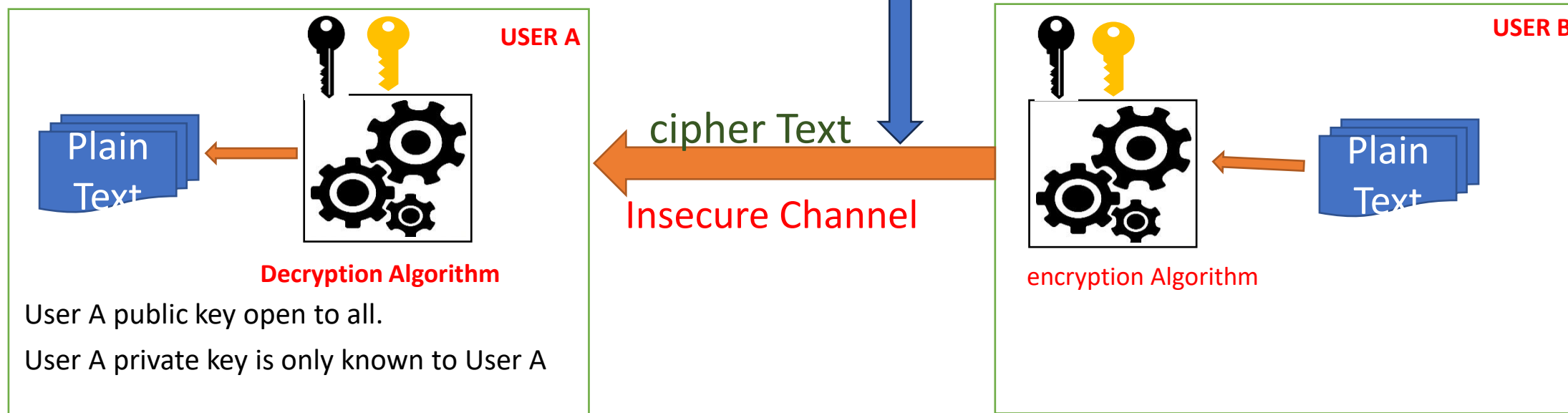
- In symmetric key cryptography, same key (One key) is used both at sender and receiver to perform encryption and decryption operations respectively. But in Asymmetric key cryptography, different keys are used at sender and receiver.
- In the figure, USER B encrypts a message using public key of USER A which can only be decrypted by USER A's private key.

Key details	A should know	B should know
A's private key	Yes	No
A's public key	Yes	Yes
B's private key	No	Yes
B's public key	Yes	Yes

USER A public key is known to hacker



USER A public key is known to USER B



Public Key Cryptography

- Public key cryptography or asymmetric cryptography is a cryptographic system that uses pair of keys.
- Public keys, which may be disseminated widely, and private keys, which are known only to the owner.
- In such a system, any person can encrypt a message using the receiver's public key, but that encrypted message can only be decrypted with the receiver's private key.